

Functional Annotation of *nadD* (Q88DL5, PP_4810) in *Pseudomonas putida* KT2440

Gene/Protein Identity Verification

- **UniProt:** Q88DL5
- **Gene:** *nadD* (Ordered locus PP_4810)
- **Protein:** Probable nicotinate-nucleotide adenylyltransferase; EC 2.7.7.18
- **Synonyms:** NaMN adenylyltransferase (NaMNAT/NAMNAT); deamido-NAD⁺ diphosphorylase/pyrophosphorylase
- **Organism:** *Pseudomonas putida* strain KT2440 (ATCC 47054 / DSM 6125)
- **Family/domains:** NadD family (HAMAP MF_00244); NadD/NMNAT (IPR005248), Cyt_trans-like (IPR004821), Rossmann-like $\alpha/\beta/\alpha$ fold (IPR014729), CTP_transf_like (PF01467)

Verification outcome: The gene symbol *nadD*, the protein description, the assigned family (NadD), and the domain architecture are all mutually consistent and match a large, coherent body of literature on the bacterial enzyme **nicotinate mononucleotide adenylyltransferase**. There is no ambiguity: *nadD* here denotes the NAD-biosynthetic enzyme, not any similarly named unrelated gene. No experimental study on the *P. putida* KT2440 ortholog specifically was found, so functional statements are inferences from (i) the strongly conserved NadD enzyme family characterized in *E. coli*, *Bacillus subtilis*, *B. anthracis*, *M. tuberculosis*, *S. pyogenes*, and humans, and (ii) the UniProt/HAMAP family assignment. These inferences are high-confidence given the diagnostic domain signatures and the deep conservation of the family.

1. Summary (Answer to the Research Question)

nadD encodes **nicotinate mononucleotide adenylyltransferase** (NaMNAT, EC 2.7.7.18), a cytoplasmic enzyme that catalyzes the **penultimate, committed step of NAD⁺ biosynthesis**: the reversible transfer of an adenylyl (AMP) group from ATP to **nicotinic acid mononucleotide (NaMN)**, producing **nicotinic acid adenine dinucleotide (NaAD, "deamido-**

NAD⁺) and **inorganic pyrophosphate (PPi)**. NaAD is subsequently amidated by NAD synthetase (NadE) to give mature NAD⁺. Bacterial NadD is **strictly selective for the deamidated substrate NaMN** (versus the amidated NMN), placing it in the deamidated NAD route that serves both de novo and salvage synthesis. The enzyme adopts a Rossmann-like nucleotidyltransferase fold, uses an ordered (ATP-first) mechanism, and is **essential and conserved across bacteria**, making it a validated antibacterial drug target.

2. Primary Function: Reaction and Substrate Specificity

Reaction catalyzed (EC 2.7.7.18):



This is the adenylation of NaMN to form NaAD, the immediate precursor of NAD⁺ (Gou et al., 2012,

P 23289308

). Knockdown studies confirm NaMN is the substrate and NaAD the product: suppressing NadD causes NaMN to accumulate while NaAD and the NAD(H) pool are depleted (Rodionova et al., 2014,

P 24549842

Substrate specificity — NaMN over NMN. Within the pyridine-nucleotide adenylyltransferase (PNAT) superfamily, substrate preference is modular: bacterial NadD is the **strictly NaMN-selective** member, in contrast to NMN-selective NadR/NadM and bifunctional mammalian NMNAT (Sorci et al., 2025,

P 41109367

). Direct biochemistry on the *B. subtilis* enzyme showed it can adenylate both NaMN and NMN in vitro but is specific for the nicotinate (carboxylated/deamidated) form, with the nicotinic-acid carboxylate recognized through active-site backbone contacts (Olland et al., 2002,

P 11704676

). Because *P. putida* NadD is assigned to the NadD family, it is predicted to share this strict NaMN preference and thus to operate in the **deamidated** branch (NaMN → NaAD → NAD⁺) rather than the amidated (NMN → NAD⁺) branch.

Co-substrate: ATP is the adenylyl donor; Mg^{2+} is the typical divalent cation cofactor for this nucleotidyltransferase chemistry.

3. Structure and Mechanism

Bacterial NadD orthologs share a **monomeric fold of the nucleotidyltransferase α/β phosphodiesterase superfamily** — a dinucleotide-binding **Rossmann-like fold** — carrying the conserved N-terminal (T/H)XGH adenylyltransferase motif that engages ATP (Olland et al., 2002,

P 11704676

B. subtilis NadD is a functional **dimer**. Steady-state kinetic analysis of *M. tuberculosis* NadD established a **sequential ordered mechanism in which ATP binds first**, enabling productive NaMN binding and catalysis, coupled to an open $\leftrightarrow$ closed active-site conformational transition (Rodionova et al., 2015,

P 25631047

The UniProt annotation of Q88DL5 (Rossmann-like $\alpha/\beta/\alpha$ fold IPR014729; CTP_transf_like PF01467; Cyt_trans-like IPR004821) is fully consistent with this architecture, supporting the mechanistic inference for the *P. putida* enzyme.

3a. Residue-Level Confirmation in the *P. putida* Ortholog

Direct analysis of the UniProt Q88DL5 sequence (230 aa) confirms the family assignment at the residue level, not merely by homology inference:

- **Catalytic (T/H)XGH motif:** the N-terminal signature **GGTFDPVHIGH** (residues 25–35) embeds the "**HIGH**" motif (His32–x–Gly34–His35), the diagnostic adenylyltransferase motif that binds ATP and positions its α -phosphate for adenylyl transfer.
- **Rossmann-like dinucleotide-binding loop:** the glycine-rich **...RRIGILGGTFDP...** region matches the phosphate-binding loop of the α/β nucleotidyltransferase fold.
- **Substrate-binding (S/T)XXXX(R/K) motif:** present in the C-terminal half (e.g., SATQIR at 194, SGKSVR at 204), matching the conserved motif described for bacterial NaMN adenylyltransferases (Olland et al., 2002,

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- **Size:** 230 aa, typical of single-domain bacterial NadD (~210–230 aa).

These signatures in the actual *P. putida* protein confirm an intact NadD catalytic apparatus.

Orthology to a validated enzyme. Global pairwise alignment of *P. putida* NadD (230 aa) against *E. coli* K-12 NadD (P0A752, 213 aa; UniProt evidence level PE=1, protein) gives **38.8% identity** (81/209 aligned columns) — well above the twilight zone for a compact single-domain enzyme — with the **identical catalytic signature** (*E. coli* GGT FDPVHYGH vs *P. putida* GGT FDPVHIGH). Since *E. coli* NadD is an experimentally characterized, essential NaMN adenyltransferase and validated inhibitor target (Sorci et al., 2009,

P 19716475

this orthology transfers experimental function to Q88DL5 with high confidence.

4. Subcellular Localization

NadD is a soluble small-molecule metabolic enzyme with **no signal peptide and no transmembrane segments**; it carries out adenyl transfer in the **bacterial cytoplasm**, where NAD biosynthesis and the NAD(H)/NADP(H) cofactor pools reside. (By contrast, the human ortholog NMNAT-1 is nuclear via a nuclear-localization signal — Zhou et al., 2002,

P 11788603

— a eukaryote-specific feature not applicable to bacterial NadD.)

5. Pathway Context in *P. putida*

NadD sits at the convergence point of NAD⁺ biosynthesis:

- **Upstream inputs to NaMN:** the de novo route (aspartate → quinolinate → NaMN via NadA/NadB/NadC) and the deamidated salvage route (nicotinamide/nicotinate → NaMN via PncA/PncB) both feed NaMN.
- **NadD step:** NaMN + ATP → NaAD + PPi.
- **Downstream:** NAD synthetase **NadE** amidates NaAD to NAD⁺.

NadD and NadE are the two indispensable downstream enzymes shared by all NAD-producing routes (Sorci et al., 2010,

P 20926389

Sorci et al., 2013, [P 23204464](#)).

In *P. putida* KT2440 this cofactor supply is especially important: the organism has an extensive aerobic, redox-intensive catabolism (mono-/dioxygenases, dehydrogenases) that continuously consumes and regenerates NAD(H)/NADP(H). Notably, *P. putida* KT2440 also possesses a well-known **nicotinate (nic) degradation locus**; intracellular nicotinic-acid pools that fluctuate with this catabolism can influence NAD biosynthetic flux (Brickman & Armstrong, 2018,

[P 29485696](#)), underscoring the role of NadD as a gatekeeper channeling nicotinate-derived NaMN into the NAD⁺ pool.

6. Evidence of Essentiality and Physiological Importance

- **Essentiality:** NadD is essential for cell viability in bacteria (Olland et al., 2002,

[P 11704676](#)); Sorci et al., 2010,

[P 20926389](#)).

- **Genetic validation:** Inducible degradation of NadD in *M. tuberculosis* is strongly **bactericidal** in both replicating and non-replicating states, causing NaMN accumulation, NAD(H) depletion, and metabolic catastrophe (Rodionova et al., 2014,

[P 24549842](#)).

- **Metabolic engineering corroboration:** Overexpression of *nadD* in *E. coli* raised total NAD(H) ~2.6-fold and rebalanced NADH/NAD⁺, restoring anaerobic growth — confirming NadD flux controls the NAD pool (Gou et al., 2012,

[P 23289308](#)).

- **Druggability/selectivity:** Structure-based inhibitors selectively inhibit bacterial NadD (*E. coli*, *B. anthracis*) with no effect on human NMNAT, and a co-crystal defined a common inhibitor-binding site (Sorci et al., 2009,

[P 19716475](#));

Huang et al., 2010,

P 20578699

7. Supported and Refuted Hypotheses

Supported (high confidence): - H1: *nadD* encodes NaMN adenylyltransferase catalyzing
NaMN + ATP → NaAD + PPi. ✓

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- H2: The enzyme is strictly NaMN-selective (deamidated pathway). ✓

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- H3: *NadD* adopts a Rossmann-like nucleotidyltransferase fold with an ordered ATP-first mechanism. ✓

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- H4: *NadD* is essential and cytoplasmic; loss depletes NAD(H). ✓

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Refuted / not applicable: - That *P. putida* *NadD* is NMN-selective or bifunctional — refuted by
strict NadD-family classification

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- Nuclear localization (a human-NMNAT feature) — not applicable to a bacterial cytoplasmic enzyme (contrast

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8. Limitations and Future Directions

- No enzymatic, structural, or knockout study exists for the *P. putida* KT2440 NadD specifically; all functional claims are inferences from orthologs and family/domain assignment (though these are among the most conserved and best-characterized in bacteria).
- Future work: recombinant expression and kinetic characterization of *P. putida* NadD (kcat/Km for NaMN, NMN, ATP), an experimental crystal or cryo-EM structure, and conditional-essentiality testing in KT2440 would confirm the inferred properties directly. Given KT2440's nicotinate-degradation capacity, quantifying how nicotinate catabolic flux versus NadD biosynthetic flux partition the NaMN/nicotinate pool would be biologically informative.

Key References

- Gou et al. 2012,

[P 23289308](#)

— reaction (NaMN→NaAD); flux control of NAD pool.

- Rodionova et al. 2014,

[P 24549842](#)

— essentiality; bactericidal knockdown; substrate/product identities.

- Rodionova et al. 2015,

[P 25631047](#)

— structure, ordered ATP-first mechanism.

- Olland et al. 2002,

[P 11704676](#)

— essentiality, NaMN specificity, Rossmann-like fold, dimer.

- Sorci et al. 2025,

[P 41109367](#)

— bacterial NadD strictly NaMN-selective.

- Sorci et al. 2010,

P 20926389

— conservation, essentiality, drug-target status.

- Sorci et al. 2013,

P 23204464

— NadD/NadE indispensable downstream NAD enzymes.

- Sorci et al. 2009,

P 19716475

Huang et al. 2010,

P 20578699

— selective bacterial inhibitors, inhibitor-binding site.

- Zhou et al. 2002,

P 11788603

— human NMNAT (contrast for specificity/localization).

- Brickman & Armstrong 2018,

P 29485696

— *P. putida* KT2440 nicotinate degradation locus and NAD biosynthesis link.